

Motivation

Computer-Use Agents (CUAs) represent an emerging paradigm in human-computer interaction in which users specify high-level goals and autonomous agents execute them by perceiving and acting directly within desktop environments. Although recent systems have made substantial progress, their reliability remains limited by the difficulty of determining whether a task has actually been completed, which in turn constrains both deployment and reinforcement-learning fine-tuning.

Most existing approaches rely on brittle benchmark scripts, hand-crafted rules, or manual inspection. These solutions do not scale well across heterogeneous applications and operating systems.

Research Questions

The present study investigates the following research questions:

- **RQ1:** Can Vision-Language Models (VLMs) act as autonomous evaluators of task completion?
- **RQ2:** Can noisy evaluator feedback be used as a reward signal for reinforcement learning fine-tuning?

The key challenge is that evaluator outputs are imperfect: false positives and false negatives can bias learning. Therefore, autonomous evaluation must be treated as a noisy reward estimation problem, not as exact supervision.

User input: "Open Apple website in Safari"

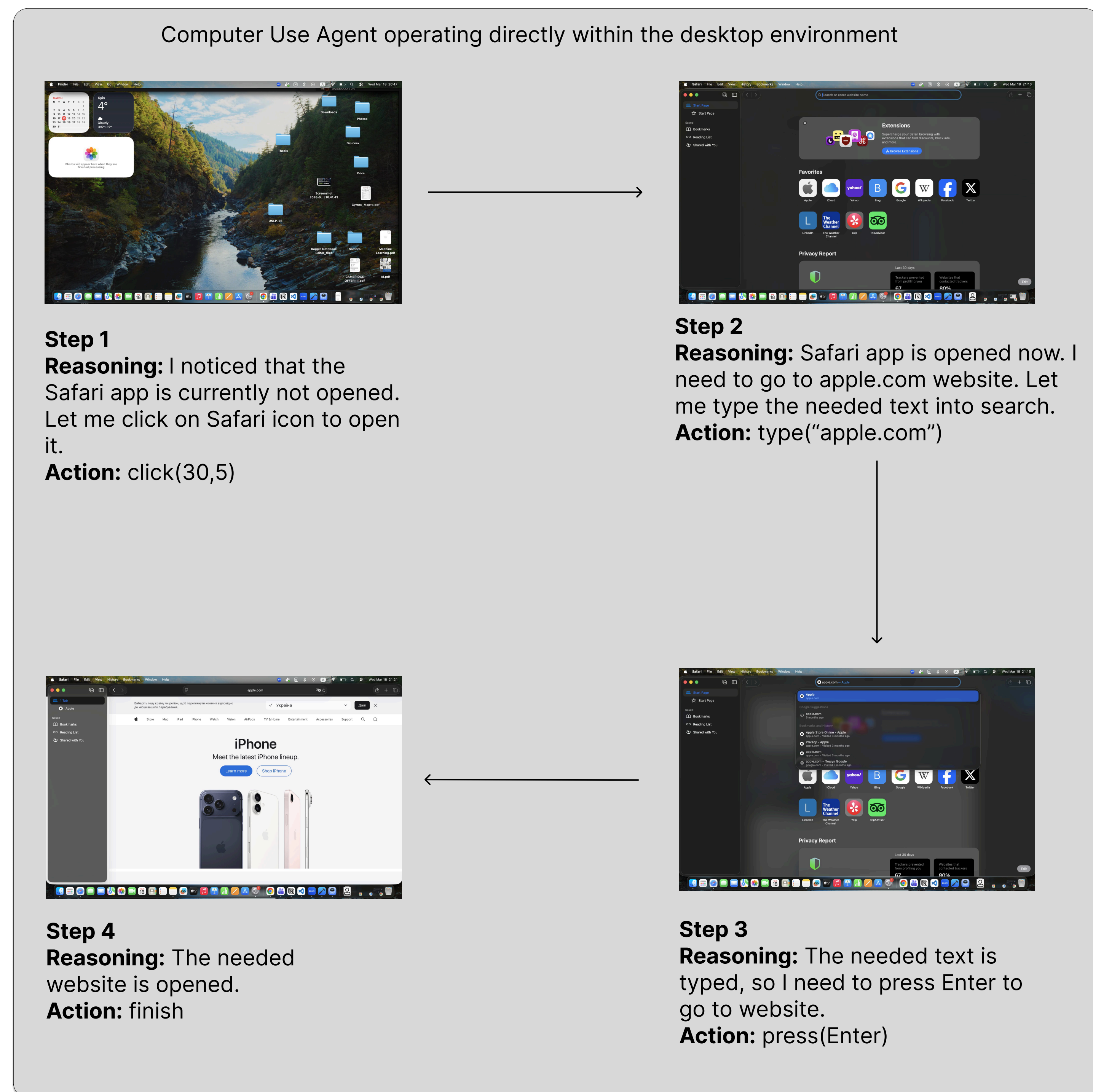


Figure 1. Illustration of a CUA operating directly within a desktop environment. Given the user instruction "Open Apple website in Safari", the agent interacts with the graphical interface through a sequence of actions, including clicking, typing, and pressing Enter, to complete the task.

Evolution of Computer Use Agents

Figure 2 shows the rapid development of CUAs in recent years, suggesting the need for reliable evaluation and learning algorithms for them.

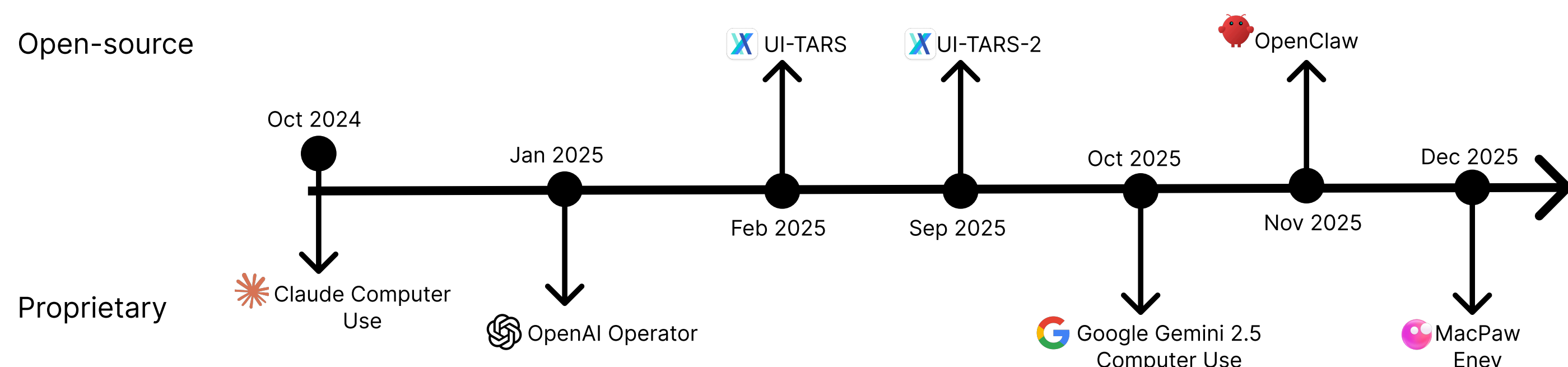


Figure 2. Recent timeline of major CUAs releases, illustrating the rapid emergence of commercial and open-source CUA systems in 2024–2025.



Study Methodology

The pipeline has two stages shown in the Figure 3:

1. **Autonomous evaluation:** a VLM judges task completion from the final GUI state.
2. **RL fine-tuning:** PPO updates the CUA using either the raw evaluator reward $\tilde{r}_i$ or the corrected reward $\hat{r}_i$.

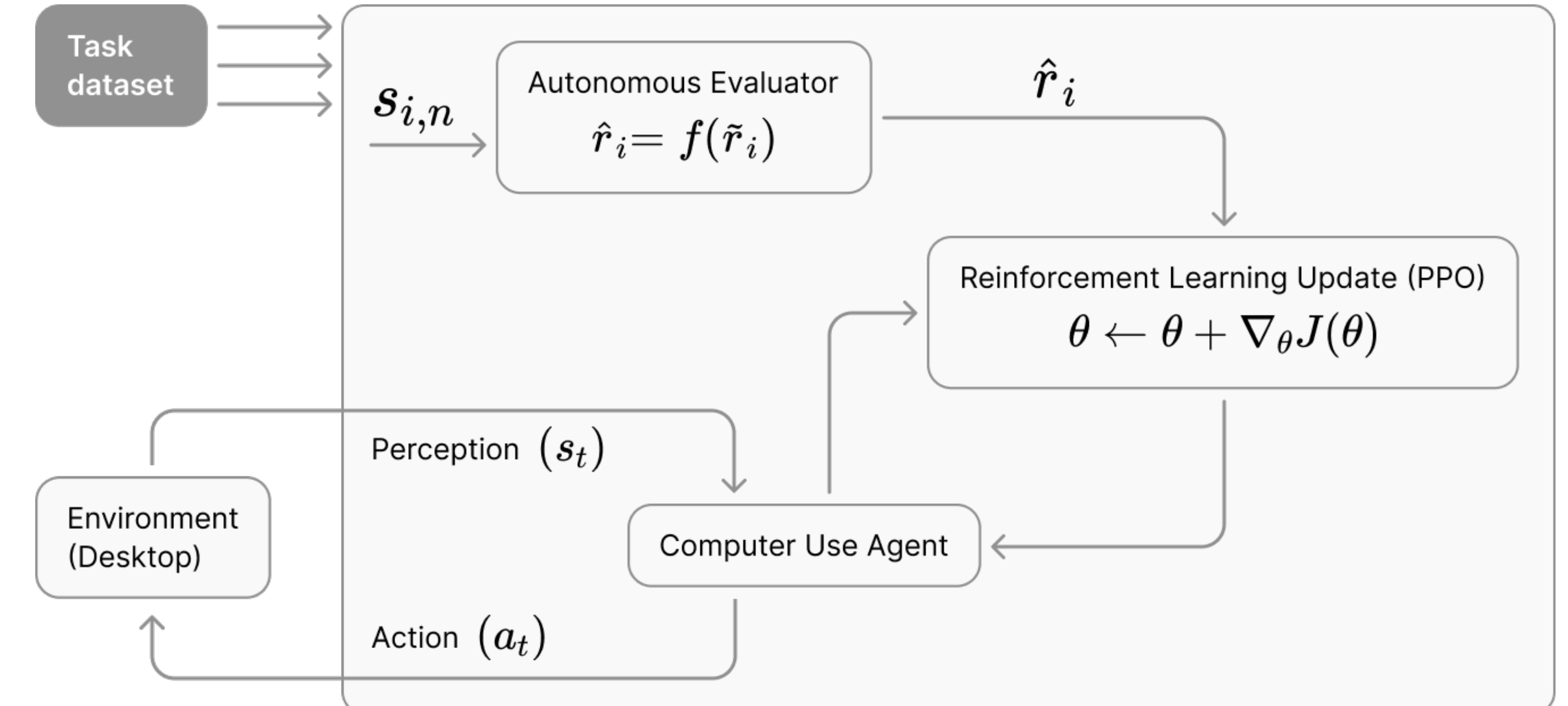


Figure 3. Overview of the reinforcement-learning pipeline with noise-corrected autonomous rewards.

Specifically, we use a VLM-based evaluator to judge whether a task is complete from:

$$(x_{i,T}, d_i),$$

where $x_{i,T}$ is the final screenshot and d_i is the task description.

The evaluator returns a noisy binary signal

$$\tilde{r}_i \in \{0, 1\}.$$

To make this usable for RL, we model evaluator noise with false-negative and false-positive rates:

$$e_+ = \Pr(\tilde{r}_i = 0 \mid r_i = 1), \quad e_- = \Pr(\tilde{r}_i = 1 \mid r_i = 0).$$

This yields a corrected reward estimator:

$$\hat{r}_i = \frac{\tilde{r}_i - e_-}{1 - e_+ - e_-}.$$

Experimental Setting

We evaluate across three desktop benchmarks:

- macOSWorld
- Windows Agent Arena
- OSWorld (Linux)

We also construct a synthetic cross-platform task dataset spanning 42 applications per OS and 2,520 tasks per OS, used for RL fine-tuning and evaluator calibration.

Results

Table 1. Zero-shot task-completion evaluation accuracy across three CUA policies.

Evaluator	Operator	Claude CU	UI-TARS
Proprietary			
GPT-4o	0.61	0.69	0.64
Claude 3.5 Sonnet	0.69	0.71	0.73
Open-source			
LLaVA-v1.5-7B	0.56	0.61	0.52
InternVL 2-8B	0.62	0.67	0.61
Qwen2-VL-7B	0.68	0.66	0.70

Observation. As shown in Table 1, VLMs can reliably assess task completion in a zero-shot setting. Claude 3.5 Sonnet achieves the strongest proprietary performance, while Qwen2-VL-7B is the strongest open-source evaluator.

Table 2. Task success under zero-shot and reinforcement-learning fine-tuning with raw versus corrected evaluator rewards.

Training setup	macOS	Win	Linux
Zero-shot baseline	0.084	0.331	0.283
RL + raw reward, unified	0.129	0.383	0.385
RL + corrected reward, unified	0.144	0.403	0.399
RL + raw reward, per-OS	0.149	0.357	0.418
RL + corrected reward, per-OS	0.203	0.442	0.432

Observation. Table 2 shows that RL fine-tuning improves task success on all three benchmarks, and that the corrected reward estimator consistently outperforms the raw evaluator signal.

Conclusions & Discussion

This thesis shows that autonomous evaluation and learning can be coupled: VLMs can judge CUA task success from visual observations, and their outputs can be transformed into a principled reward signal for reinforcement learning.

The central result is that correcting evaluator noise improves policy learning. In the per-OS setting, the proposed approach achieves the strongest performance across all three benchmarks.